161. (e) Photochemical chlorination of alkane take place by free radical mechanism which are possible by Homolysis of $C - C$ bond

$$Cl_2 \xrightarrow{hv} Cl^{\bullet} + Cl^{\bullet}$$

$$CH_3 - CH_3 + Cl^{\bullet} \rightarrow CH_3Cl^{\bullet} + \dot{C}H_3$$

162. **(c) Gasoline** — The 70 – 200° C fraction containing $C_6 - C_{10}$ hydrocarbons corresponds to gasoline (petrol).

163. (a) Producer gas – CO and N_2

164. (c) Among alkanes, boiling point increase with increasing molecular weight. For isomeric alkanes straight chain alkanes have higher boiling point than the branched alkanes.

165. (b) Graphite is a good conductor of heat of electricity.

166. (d) Among the isomeric alkanes, the normal isomer has a higher boiling point than the branched chain isomer. The greater the branching of the chain, the lower is the boiling point. The *n*-alkane have larger surface area in comparison to branched chain isomer (as the shape approaches that of a sphere in the branched chain isomers). Thus, intermolecular forces are weaker in branched chain isomers, there fore they have lower point in comparision to straight chain isomers.

167. (d) The octane numbers of Fuel can be improved by increasing the percentage of branched chain alkanes, alkenes and aromatic hydrocarbon. Thus octane number can be changed by isomerisation (reforming), alkylation and aromatisation (cyclisation) etc.

168. (c) The approximate composition of gasoline is $C_6 - C_{11}$ at boiling point 70-200° C and is used in motor fuel, dry cleaning, petrol gas etc.

169. (c) $CH_4 + O_2 \rightarrow CO_2 + 2H_2O$

170. **(c)** Straight chain olefins has highest knocking.

171. (a) $Al_4C_3 + 12H_2O \rightarrow 3CH_4 + 4Al(OH)_3$

$$CaC_2 + 2H_2O \rightarrow C_2H_2 + Ca(OH)_2$$

172. (d) Except 2,2 dimethyl butane rest compound contain 5 carbon i.e., pantane while 2,2 dimethyl butane contain 6 carbon i.e., Hexane

$$CH_3-CH_2-\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}}-CH_3$$

173. (b) $$H_3C-\underset{\underset{CH_3}{|}}{CH}-CH_2-CH_3+Br_2 \longrightarrow H_3C-\overset{\overset{Br}{|}}{\underset{\underset{CH_3}{|}}{C}}-CH_2-CH_3$$ Major

174. (b) $$H_3C-\overset{\overset{CH_3}{|}}{\underset{\underset{H}{|}}{C}}-\overset{\overset{CH_3}{|}}{\underset{\underset{H}{|}}{C}}-CH_3 \xrightarrow{Cl_2} H_3C-\overset{\overset{CH_2Cl}{|}}{\underset{\underset{H}{|}}{C}}-\overset{\overset{CH_3}{|}}{\underset{\underset{H}{|}}{C}}-CH_3 + H_3C-\overset{\overset{CH_3}{|}}{\underset{\underset{Cl}{|}}{C}}-\overset{\overset{CH_3}{|}}{\underset{\underset{H}{|}}{C}}-CH_3$$

175. (c) $$C_2H_5Cl+H_2 \xrightarrow{Pd/C} C_2H_6+HCl$$

This reaction is used for the preparation of pure alkanes.